

NIH Research Project Grant (R01)

National Institute of Neurological Disorders and Stroke (NINDS)

Grant Number: 1R01NS123456-01
FOA Number: PA-23-152
Study Section: Neurotransporters, Receptors, and Calcium Signaling (NTRC)
Funding Mechanism: R01 - Research Project Grant
Submission Date: February 5, 2024

Project Information

Project Title: Neural Circuit Mechanisms of Working Memory in Prefrontal Cortex
Principal Investigator: Dr. James Park
Degree: Ph.D.
Institution: University of Idaho
Department: Department of Biological Sciences - Neuroscience Program
Total Budget: \$1,250,000
Project Period: July 1, 2024 - June 30, 2029
Sponsoring Agency: National Institutes of Health
Human Subjects: No
Vertebrate Animals: Yes - Mouse model (Mus musculus)
Clinical Trial: No

Specific Aims

Aim 1: Characterize the firing patterns of prefrontal cortex (PFC) neurons during working memory tasks using two-photon calcium imaging in head-fixed mice.

Aim 2: Determine the causal role of PFC-to-hippocampal projections in working memory maintenance using optogenetic manipulation.

Aim 3: Develop a computational model of PFC circuit dynamics that predicts working memory capacity from neural population activity patterns.

Key Personnel

Name	Role	Effort	Department
Dr. James Park	PI	30%	Biological Sciences
Dr. Lisa Yamamoto	Co-Investigator	15%	Psychology
Dr. Raj Patel	Co-Investigator	10%	Computer Science
TBD	Postdoctoral Fellow	100%	Biological Sciences
TBD	Graduate Student 1	50%	Neuroscience Program
TBD	Graduate Student 2	50%	Neuroscience Program

Budget Justification

Category	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Personnel	\$120,000	\$124,000	\$128,000	\$132,000	\$121,000	\$625,000
Equipment	\$85,000	\$15,000	\$10,000	\$10,000	\$5,000	\$125,000
Travel	\$8,000	\$10,000	\$10,000	\$12,000	\$10,000	\$50,000
Supplies	\$30,000	\$28,000	\$28,000	\$25,000	\$24,000	\$135,000
Other	\$10,000	\$10,000	\$10,000	\$10,000	\$10,000	\$50,000
Indirect (52%)	\$51,000	\$53,000	\$55,000	\$57,000	\$49,000	\$265,000
Total	\$304,000	\$240,000	\$241,000	\$246,000	\$219,000	\$1,250,000

Research Strategy

Working memory, the ability to transiently hold and manipulate information, is fundamental to cognition. Deficits in working memory are observed in schizophrenia, ADHD, and age-related cognitive decline. Despite decades of research, the circuit-level mechanisms by which prefrontal cortex (PFC) neurons maintain persistent activity during working memory remain poorly understood.

This project employs state-of-the-art two-photon calcium imaging to record the activity of hundreds of PFC neurons simultaneously in behaving mice performing a delayed non-match-to-sample task. Combined with optogenetic manipulation of specific projection pathways and computational modeling, we aim to provide a comprehensive understanding of how PFC circuits support working memory.

Vertebrate Animals

All procedures will be performed in accordance with the NIH Guide for the Care and Use of Laboratory Animals and approved by the University of Idaho IACUC (Protocol #2024-015). We will use approximately 200 adult C57BL/6J mice (both sexes, 8-12 weeks old) over the project period. Mice will be housed in the university vivarium with standard 12h light/dark cycles and ad libitum food and water access.